

Entities and Attributes

- Driver
 - Driver ID (primary key)
 - First Name
 - Last Name
 - Driver Number
 - Current Team (foreign key)
- Team
 - Team ID (primary key)
 - Team Name
 - Drivers (derived attribute)
- Race
 - Race ID (primary key)
 - Location
 - Track Name (foreign key)
 - Race Dates
 - Race Time (UTC)
 - Round Number
 - Has Sprint
- Track
 - Track ID (primary key)
 - Track name
 - Turns
 - Elevation
 - Top Speed
 - Lap Record
- Qualifying
 - Race ID (primary key)
 - Driver ID (primary key)
 - Position
- Result
 - Race ID (primary key)
 - Driver ID (primary key)
 - Starting Position
 - Finishing Position
 - Points
- Lap
 - Race ID (primary key)
 - Driver ID (primary key)
 - Lap number
 - Lap Time
 - Tire

Relationships

- Driver DRIVES_FOR Team → 1:1 (Each driver drives for only one team at a time) → Total:Partial (Every driver must drive for a team and a team may exist without drivers)
- Team SIGNS Driver → 1:N (One team has many drivers) → Partial:Total
- Track HOSTS Race → 1:N (One track hosts many races) → Partial:Total (Not every track hosts a race, but a race must occur on a track)
- Race HAS Qualifying → 1:N (Each race has three qualification rounds) → Total:Total (Every race has a qualifying session, and each qualifying sessions belongs to a race)
- Driver QUALIFIES_IN Race → 1:N (One driver qualifies in many races) → Partial:Total (Not all drivers participate in every race, but all races have drivers in qualifying)
- Race HAS Result → 1:N (One race has many results) → Total:Total (Each race has a result, and each result belongs to a race)
- Driver ACHIEVES Result → 1:N (One Driver has many race results) → Partial:Total (Driver may not attend all races, but all results must belong to a driver)
- Race HAS Lap → 1:N (Each race has many laps) → Total:Total (A race must consist of at least one lap, and each lap belongs to a race)
- Driver COMPLETES Lap → 1:N (Each driver completes many laps) → Partial:Total (Drivers may not complete laps (DNS/DNF), but every lap entry must belong to a driver)

Derived

- Driver EARNs Points
 - Calculate from Result
- Team ACCUMULATES Points
 - Sum of driver points

Phase 1:

- Live season status
- F1 Fantasy aid
- Project race winner
 - PowerBI
- Weather at the track